

Assignment 5: ADDITION OF NEW VOLUME OUTPUT

The objective of this assignment is to enhance the simulation by incorporating an additional volume output variable and analyzing its spatial distribution within the computational domain.

Implementation:

The implementation was carried out in two steps: configuration-level modification and code-level integration.

Configuration Changes:

The following parameters were added in the configuration file to enable the required output:

- VOLUME_OUTPUT = (PRIMITIVE, SPEEDSOUND)
- HISTORY_OUTPUT = (RMS_RES, AVG_SOUNDSPEED)
- SCREEN_OUTPUT = (INNER_ITER, RMS_DENSITY, AVG_SOUNDSPEED)

The speed of sound variable was incorporated into the output module, and the code was rebuilt to enable its availability in the simulation outputs.

Simulation Setup:

The simulation corresponds to a steady-state axisymmetric turbulent jet case using the SST turbulence model.

Key parameters used in the simulation are:

- Mach number: 0.12
- Reynolds number: 50,000
- Freestream temperature: 300 K
- Fluid model: Ideal gas
- Turbulence model: SST

The boundary conditions include inlet with prescribed flow properties, outlet with static pressure, symmetry along the axis, and isothermal wall condition.

Results:

The output shows the spatial distribution of the added variable across the computational domain. The variation is observed across the flow field, with relatively uniform values in the freestream region and localized changes near the jet and wall regions.

Conclusion:

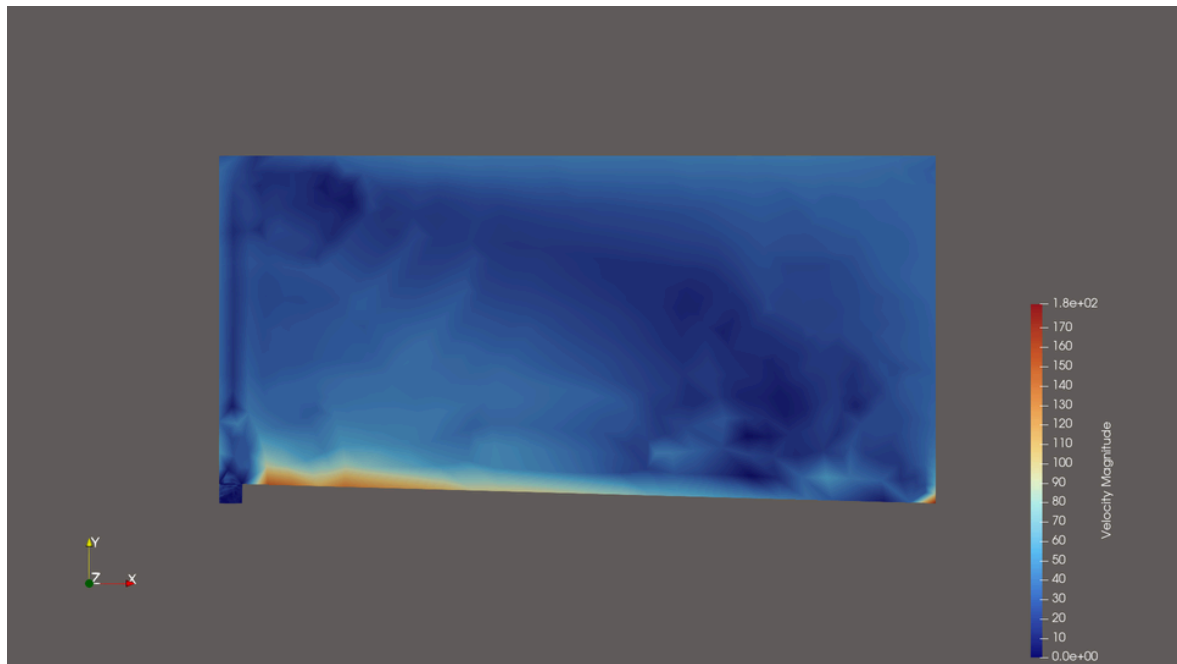
The new volume output variable was successfully integrated into the simulation through configuration and code modifications. The generated results demonstrate its distribution across the domain, confirming correct inclusion of the output in the simulation.

Outputs:

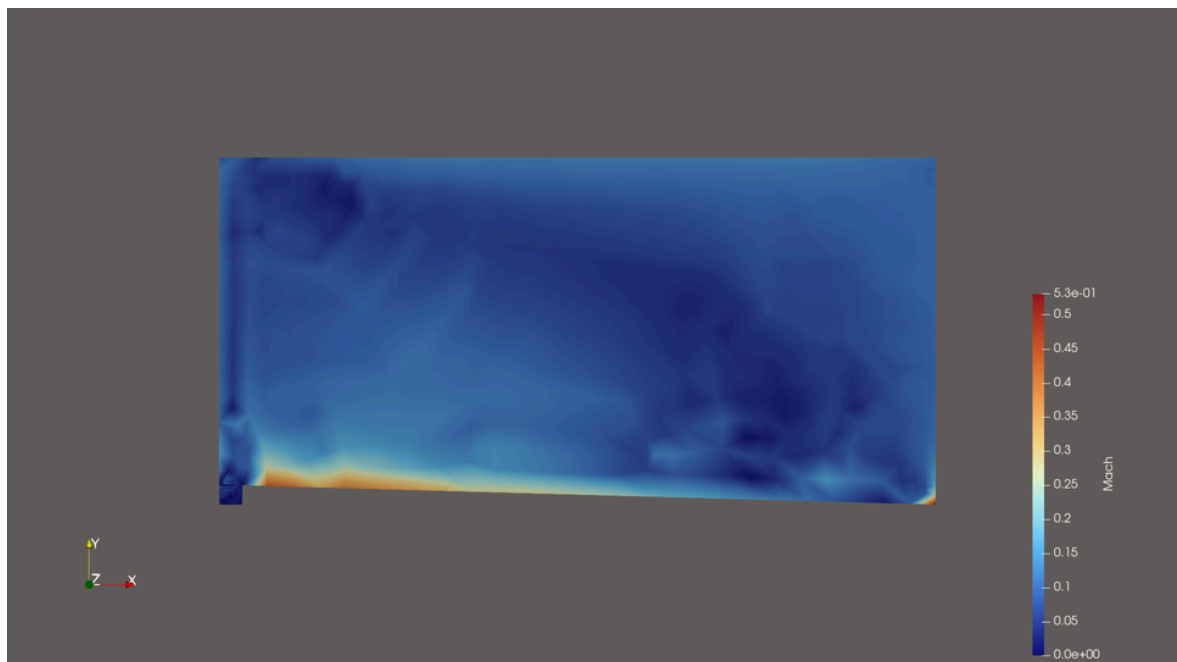


Speed of Sound

Contour of speed of sound showing its distribution in the computational domain.



Velocity Output



Mach Value Output